

Insert Arrangements

TX Series • MIL-DTL-38999 Series III Style Connectors

Selecting Your Insert Arrangement

Below is a chart that represents every available shell and insert arrangement within the series. To choose the proper insert arrangement, you must first distinguish your application requirements for contact size and quantity.

Each insert arrangement has a unique number to identify its size and contact arrangement. The first number is the shell size. The second number is the contact arrangement. It does not, however, always equal the total number of contacts. ■

Insert Arrangement Selection

Insert Arrangement	Military Shell Size	Total Contacts	Hermetic Available	Service Rating	Contact Size						
					#22D	#20	#16	#12	#12 Coax ¹	#8 Coax ²	#8 Twinax ³
9-35	A	6	✓	M	6						
9-94 [†]	A	2		M		2					
9-98	A	3	✓	I		3					
11-2	B	2	✓	I			2				
11-5	B	5	✓	I		5					
11-35	B	13	✓	M	13						
11-54	B	4		II	4						
11-98	B	6	✓	I		6					
11-99	B	7	✓	I		7					
13-4	C	4	✓	I			4				
13-8	C	8	✓	I		8					
13-13 ^{††}	C	4		I, Fiber			2	2			
13-35	C	22	✓	M	22						
13-98	C	10	✓	I		10					
15-4 [†]	D	4		I				4			
15-5	D	5	✓	II			5				
15-15	D	15	✓	I		14	1				
15-18	D	18	✓	I		18					
15-19	D	19	✓	I		19					
15-35	D	37	✓	M	37						
15-97	D	12	✓	I		8	4				
17-2	E	39		M	38						1
17-6	E	6	✓	I				6			
17-8	E	8	✓	II			8				
17-22 [†]	E	4		Coax					2	2	
17-26	E	26	✓	I		26					
17-35	E	55	✓	M	55						
17-99	E	23	✓	I		21	2				
19-11	F	11	✓	II			11				
19-18	F	18		M	14						4
19-28	F	28	✓	I		26	2				
19-31 [†]	F	15		M	12			1		2	
19-32	F	32	✓	I		32					
19-35	F	66	✓	M	66						
21-11	G	11	✓	I				11			
21-16	G	16	✓	II			16				
21-29	G	27		I		19	4	4			
21-35	G	79	✓	M	79						
21-39	G	39	✓	I		37	2				
21-41	G	41	✓	I		41					
21-75	G	4		M							4

(Continued on next page)

Rev. 2235

Insert Arrangement Selection (Continued from previous page)

Insert Arrangement	Military Shell Size	Total Contacts	Hermetic Available	Service Rating	Contact Size						
					#22D	#20	#16	#12	#12 Coax ¹	#8 Coax ²	#8 Twinax ³
21-79 [†]	G	19		II	17					2	
23-6 [†]	H	6		M							6
23-14 [†]	H	14		I				14			
23-21	H	21	✓	II			21				
23-35	H	100	✓	M	100						
23-53	H	53	✓	I		53					
23-54 [†]	H	53		M	40		9	4			
23-55	H	55	✓	I		55					
25-4	J	56	✓	I		48	8				
25-7	J	99		Twinax	97						2
25-8	J	8		Twinax							8
25-17 [†]	J	42		M	36						6
25-19	J	19	✓	I				19			
25-20 ⁵	J	30		N		10	13		4		3
25-24	J	24	✓	I			12	12			
25-29	J	29	✓	I			29				
25-35	J	128	✓	M	128						
25-41 [†]	J	41		N, Inst.	22	3	11		2		3
25-43	J	43	✓	I		23	20				
25-46	J	46		I		40	4			2	
25-61	J	61	✓	I		61					
25-90	J	46		I		40	4				2

¹ Includes the following #12 coax contacts: TXPX12-006 (pin) or TXSX12-007 (socket) contacts, see p. B-11 for more information. † Special order contact arrangement.

² Includes the following #8 coax contacts: TXPX08-012 (pin) or TXSX08-013 (socket) contacts, see p. B-11 for more information.

³ Includes the following #8 twinax contacts: TXPW08-002 (pin) or TXSW08-003 (socket) contacts, see p. B-11 for more information.

⁴ 13-13 insert is designed to accept 2 of our standard size #16 fiber optic contacts in place of #16 power contacts, see p. B-12 for more information.

⁵ 25-20 insert is designed for MIL-STD-1760 applications and includes the following contacts:

2 x #12 coax contacts: TXPX12-006 (pin) or TXSX12-007 (socket) contacts

2 x #12 coax contacts (matched impedance): TXPX12-010 (pin) or TXSX12-011 (socket) contacts

3 x #8 twinax contacts: TXPW08-002 (pin) or TXSW08-003 (socket) contacts

HIGH-FREQUENCY CONTACTS FOR HIGH-SPEED TRANSMISSION

TX00 wall mount with eight concentric twinax contacts for high-speed data transmission in a rugged connector.



Contacts for Advanced Applications

Milnec's high-frequency contacts are the ideal choice for advanced applications that require superior electrical performance along with shielding to protect from outside electrical interference.

- High-speed RF contacts
- Fiber optic contacts
- PC tail contacts



Insert Arrangement Drawings

TX Series • MIL-DTL-38999 Series III Style Connectors

Front face of pin insert shown

Arrangement

Service Rating

Number of Contacts



9-35

M

6 x #22D



9-94

M

2 x #20



9-98

I

3 x #20



11-2

I

2 x #16



11-5

I

5 x #20



11-35

M

13 x #22D



11-54

II

4 x #22D



11-98

I

6 x #20



11-99

I

7 x #20



13-4

I

4 x #16

Arrangement

Service Rating

Number of Contacts



13-8

I

8 x #20



13-13

I, Fiber Optic

2 x #16

2 x #12



13-35

M

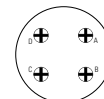
22 x #22D



13-98

I

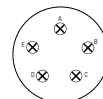
10 x #20



15-4

I

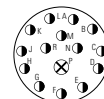
4 x #12



15-5

II

5 x #16

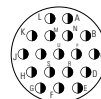


15-15

I

14 x #20

1 x #16



15-18

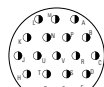
I

18 x #20

Arrangement

Service Rating

Number of Contacts



15-19

I

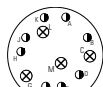
19 x #20



15-35

M

37 x #22D



15-97

I

8 x #20

4 x #16

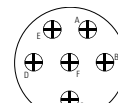


17-2

M

38 x #22D

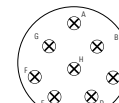
1 x #8 Twinax



17-6

I

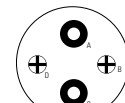
6 x #12



17-8

II

8 x #16



17-22

Coax

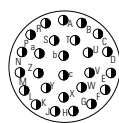
2 x #12 Coax

2 x #8 Coax

Arrangement

Service Rating

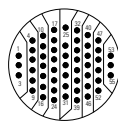
Number of Contacts



17-26

I

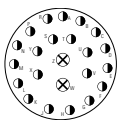
26 x #20



17-35

M

55 x #22D

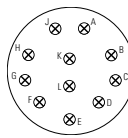


17-99

I

21 x #20

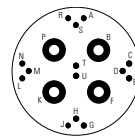
2 x #16



19-11

II

11 x #16

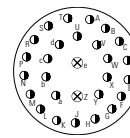


19-18

M

14 x #22D

4 x #8 Twinax

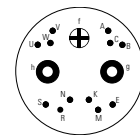


19-28

I

26 x #20

2 x #16



19-31

M

2 x #8 Coax

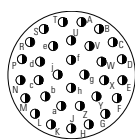
1 x #12

12 x #22D

Arrangement

Service Rating

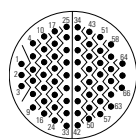
Number of Contacts



19-32

I

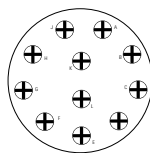
32 x #20



19-35

M

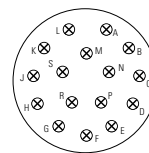
66 x #22D



21-11

I

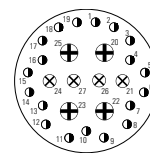
11 x #12



21-16

II

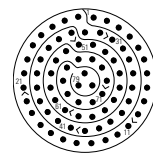
16 x #16



21-29

I

19 x #20
4 x #16, 4 x #12



21-35

M

79 x #22D

Contact Legend



8



10



12



#16



#20



#22D

(Continued on next page)

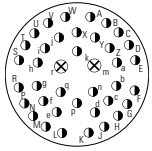
Rev. 2235

Insert Arrangement Drawings

TX Series • MIL-DTL-38999 Series III Style Connectors

(Continued from previous page)

Front face of pin insert shown

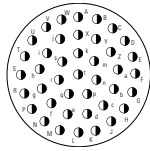


21-39

I

37 x #20

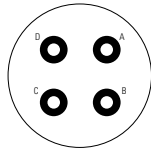
2 x #16



21-41

I

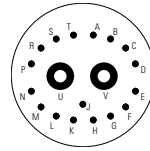
41 x #20



21-75

M

4 x #8 Twinax

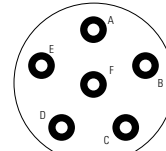


21-79

II

17 x #22D

2 x #8 Coax



23-6

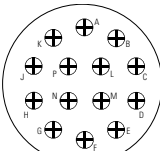
M

6 x #8 Twinax

Arrangement

Service Rating

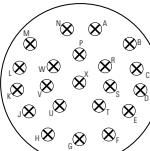
Number of Contacts



23-14

I

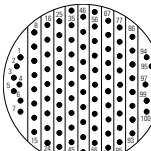
14 x #12



23-21

II

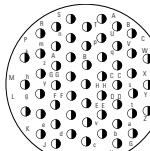
21 x #16



23-35

M

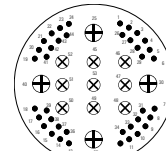
100 x #22D



23-53

I

53 x #20



23-54

M

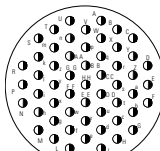
40 x #22D, 9 x #16

4 x #12

Arrangement

Service Rating

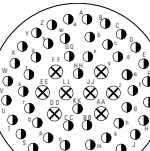
Number of Contacts



23-55

I

55 x #20

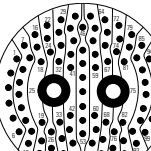


25-4

I

48 x #20

8 x #16

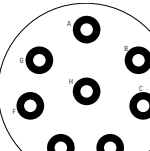


25-7

Twinax

97 x #22D

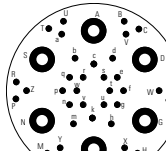
2 x #8 Twinax



25-8

Twinax

8 x #8 Twinax



25-17

M

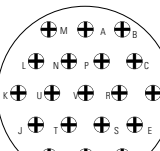
36 x #22D

6 x #8 Twinax

Arrangement

Service Rating

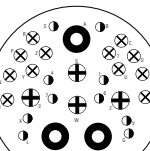
Number of Contacts



25-19

I

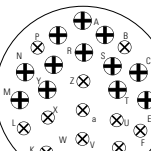
19 x #12



25-20

N

10 x #20, 13 x #16,
3 x #8 Twinax, 4 x #12 Coax

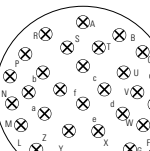


25-24

I

12 x #16

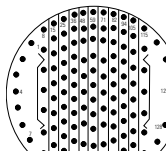
12 x #12



25-29

I

29 x #16



25-35

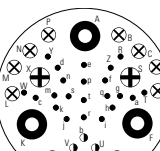
M

128 x #22D

Arrangement

Service Rating

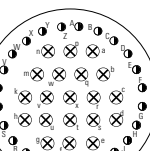
Number of Contacts



25-41

N/Inst.

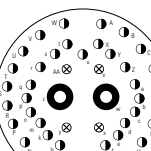
22 x #22D, 3 x #20, 11 x #16
2 x #12 Coax, 3 x #8 Twinax



25-43

I

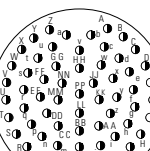
23 x #20
20 x #16



25-46

I

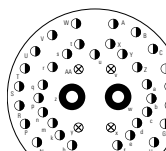
40 x #20; 4 x #16
2 x #8 Coax



25-61

I

61 x #20



25-90

I

40 x #20, 4 x #16
2 x #8 Twinax

Arrangement

Service Rating

Number of Contacts

Rev. 2235